

PH101: 2022-2023

Quantum Mechanics

Lecture 5

Arunansu Sil
Subhaditya Bhattacharya

Recap

- Wave functions to describe quantum particles satisfy Schrodinger equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x) \psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

- The wave functions must be square integrable, so that it represents the probability of finding a particle:

$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1$$

- Any act of measurement requires an operator to act on the wave function to predict outcome.

Example: $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$

- Expectation or average outcome of a measurement can be found out by:

$$\langle Q(x, p) \rangle = \int \Psi^* Q(x, \frac{\hbar}{i} \frac{\partial}{\partial x}) \Psi dx.$$

Probability current density

Recall,
probability
density:

$$\int_{-\infty}^{\infty} dx P(x, t) = \int_{-\infty}^{\infty} dx |\psi(x, t)|^2 = 1$$

This requires the wave
function to drop faster:

$$\int_{-\infty}^{\infty} dx |\psi(x, 0)|^2 < \infty$$

The complex conjugate of Schrodinger Equation:

$$-i\hbar \frac{\partial \psi^*(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*(x, t)}{\partial x^2} + V(x) \psi^*(x, t)$$

Therefore,

$$\begin{aligned} \frac{\partial}{\partial t} P(x, t) &= \frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t} \\ &= \frac{1}{i\hbar} \left(\frac{\hbar^2}{2m} \right) \left(\frac{\partial^2 \psi^*}{\partial x^2} \psi - \psi^* \frac{\partial^2 \psi}{\partial x^2} \right) \\ &= -\frac{\partial}{\partial x} \left[\frac{\hbar}{2im} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right) \right] \end{aligned}$$

Replace from
Scrodinger's
Equation.

$$\frac{\partial}{\partial t} P(x, t) + \frac{\partial}{\partial x} j(x, t) = 0$$

Continuity
Equation

Probability Current Density
or Flux

$$j(x, t) = \frac{\hbar}{2im} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right)$$

$$\begin{aligned} \frac{\partial}{\partial t} \int_a^b dx P(x, t) &= - \int_a^b dx \frac{\partial}{\partial x} j(x, t) \\ &= j(a, t) - j(b, t) \end{aligned}$$

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} dx P(x, t) = - \int_{-\infty}^{\infty} dx \frac{\partial}{\partial x} j(x, t) = 0$$

This is a
conservation law. It
expresses the fact
that the change in
the density in a
region is
compensated by
net change in flux
into that region.

Schrodinger Equation in one dimension

A quantum particle in potential $V(x)$ solves Schrodinger equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x) \Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

Second order partial differential equation.

Wave function is a function of both space and time.

We assume the potential $V(x)$ to be independent of time.

We need to find solutions to this equation

Separation of variables

We look for solutions that are of the form :

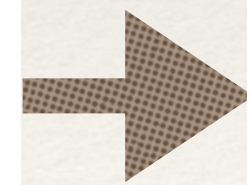
$$\Psi(x, t) = \psi(x) f(t)$$

Function of x Function of t

It may look absurd at this moment and we may think that it will only provide a tiny subset of all the solutions. But hang on ! We will be able to patch up together the separable solutions in such a way as to construct the most general solution.

Schrodinger Equation

$$\frac{\partial \Psi}{\partial t} = \psi \frac{df}{dt}, \quad \frac{\partial^2 \Psi}{\partial x^2} = \frac{d^2 \psi}{dx^2} f$$



$$i\hbar \psi \frac{df}{dt} = -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} f + V \psi f.$$

Separation of variables

Dividing both sides by $\psi(x)f(t)$

$$i\hbar \frac{1}{f} \frac{df}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V.$$

Function of t

Function of x

The only way this can possibly be true if both sides are equal to a constant E

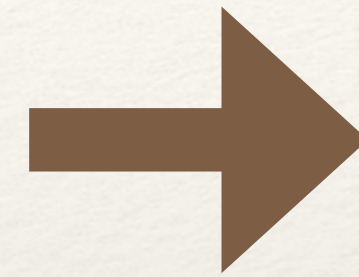
$$i\hbar \frac{1}{f} \frac{df}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V. = E$$

Time independent Schrodinger Equation

Time dependent
part:

$$i\hbar \frac{1}{f} \frac{df}{dt} = E,$$

$$\frac{df}{dt} = -\frac{iE}{\hbar} f,$$



$$f(t) = e^{-iEt/\hbar}.$$

Solution to the time dependent part

Space dependent
part:

$$-\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V = E,$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi.$$

Separation of variables has turned a partial differential equation into two ordinary differential equations.

This is called time independent Schrodinger Equation

We can't proceed further unless V is specified.

Stationary states

Separable
solution:

$$\Psi(x, t) = \psi(x)e^{-iEt/\hbar},$$

Although wave function is time dependent the probability density as well as expectation values are time independent

$$|\Psi(x, t)|^2 = \Psi^* \Psi = \psi^* e^{+iEt/\hbar} \psi e^{-iEt/\hbar} = |\psi(x)|^2$$

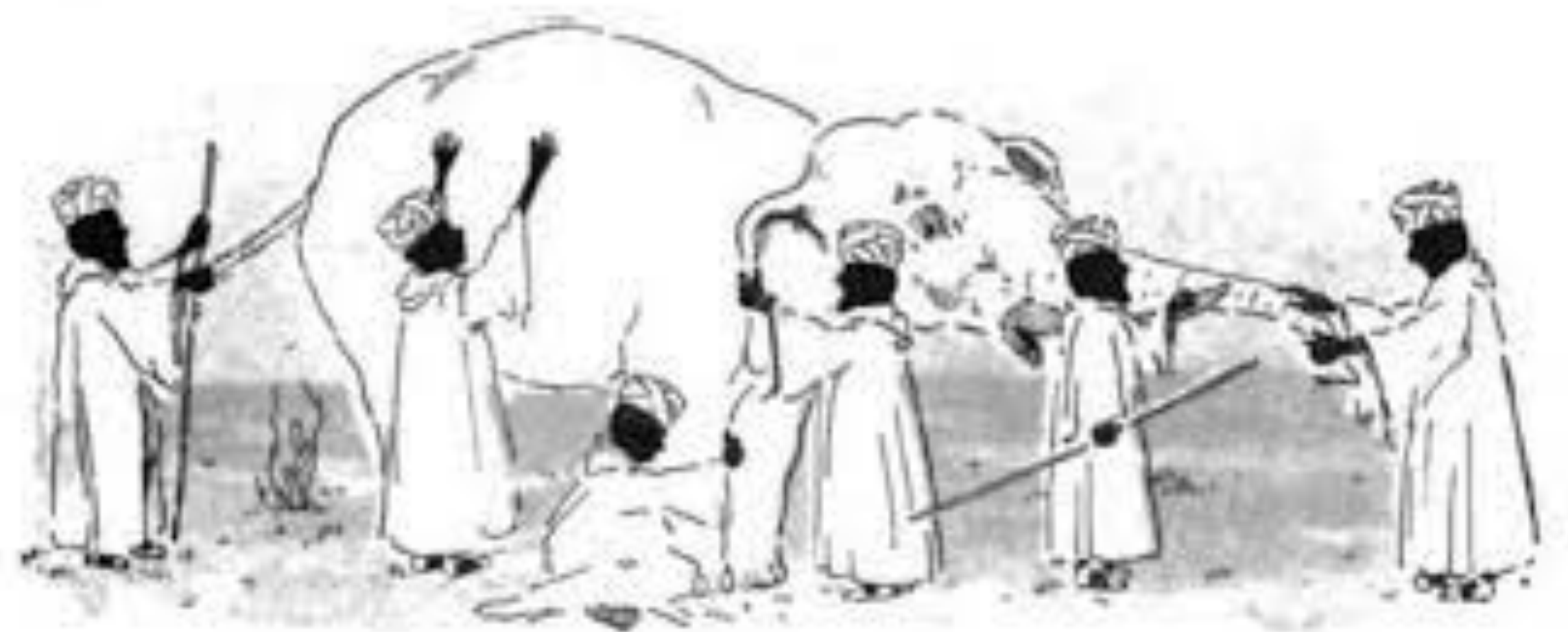
Probability density is
time independent

$$\langle Q(x, p) \rangle = \int \psi^* Q(x, \frac{\hbar}{i} \frac{d}{dx}) \psi dx.$$

$$\neq f(t)$$

Expectation value
of any dynamic
quantity is also
time independent

These states are called
stationary states



Concept of Eigenvalues and Eigenvectors

In classical mechanics, the total energy, Kinetic plus potential energy is called the **Hamiltonian**

$$H(x, p) = \frac{p^2}{2m} + V(x).$$

The Hamiltonian operator is obtained by replacing

$$\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x).$$

Thus time independent Schrodinger equation can be written as:

$$\hat{H}\psi = E\psi$$

This is called an eigenvalue equation

The values of E that solves the equation are called Eigenvalues.
The wavefunction ψ that solves the equation are called Eigenfunctions.

The expectation value of the Hamiltonian operator:

$$\langle H \rangle = \int \psi^* \hat{H} \psi dx = E \int |\psi|^2 dx = E.$$

Therefore the constant E is nothing but the energy of the state

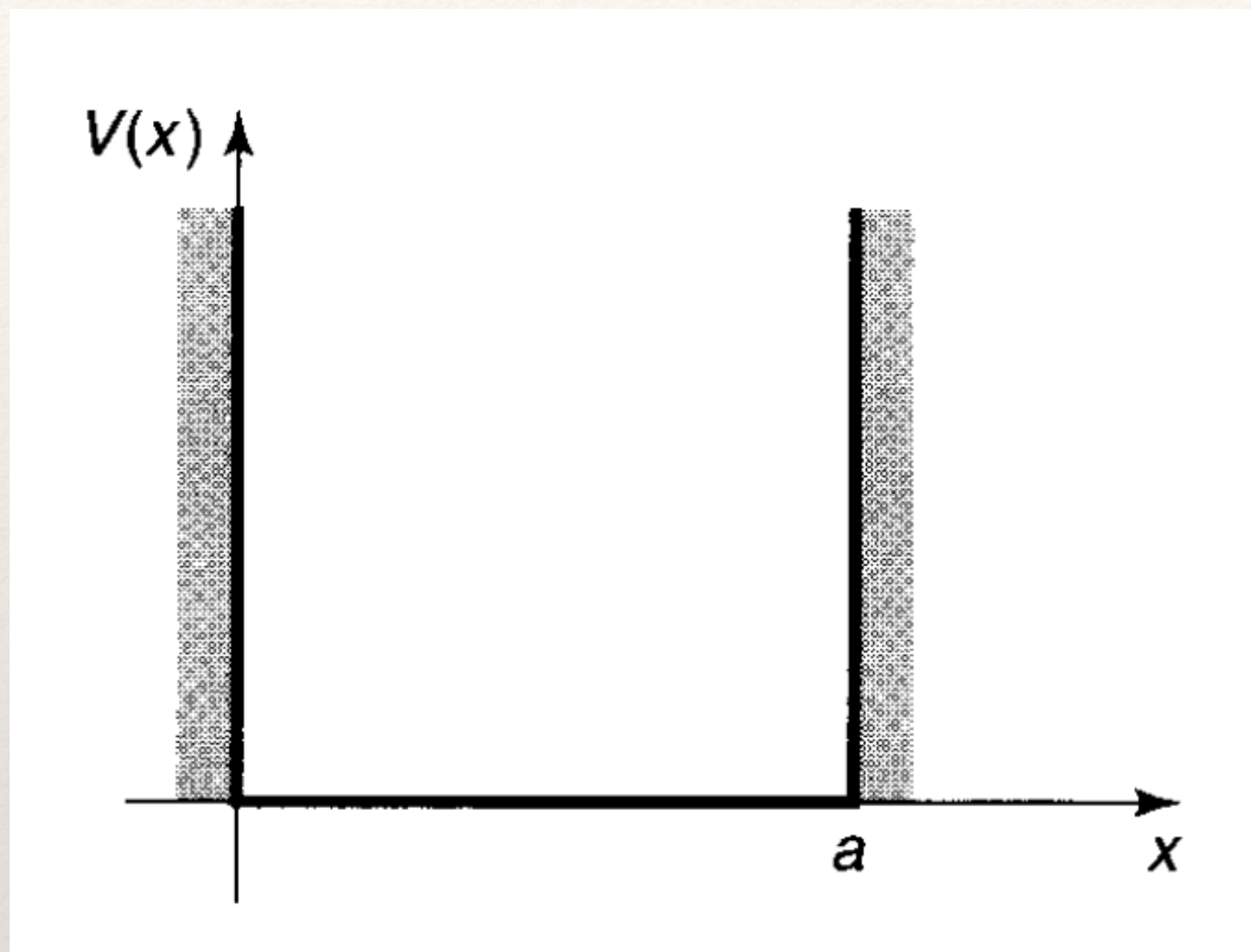
$$\hat{H}^2 \psi = \hat{H}(\hat{H} \psi) = \hat{H}(E \psi) = E(\hat{H} \psi) = E^2 \psi,$$

$$\langle H^2 \rangle = \int \psi^* \hat{H}^2 \psi dx = E^2 \int |\psi|^2 dx = E^2.$$

$$\sigma_H^2 = \langle H^2 \rangle - \langle H \rangle^2 = E^2 - E^2 = 0.$$

Every measurement of the total energy is certain to return the value E :
The energy of the stationary states have no uncertainty

Example 1: Infinite square well potential



This is also called particle in a box problem

$$V(x) = \begin{cases} 0, & \text{if } 0 \leq x \leq a, \\ \infty, & \text{otherwise} \end{cases}$$

One of the simplest examples of a bound system

- The particle is free inside, while the rigid wall prevents it to go outside.

$$\psi(x) = 0$$

At and beyond $x = \{0, a\}$

Time independent Schrodinger Equation

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi, \quad 0 \leq x \leq a$$

(Assuming $E \geq 0$) →

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad \text{where } k \equiv \frac{\sqrt{2mE}}{\hbar}.$$

Simple Harmonic Equation

The unknown constants can only be found from the boundary conditions.

Solution: $\psi(x) = A \sin kx + B \cos kx,$

Boundary Conditions and Solution

Usually, both $\psi(x)$ and $\frac{d\psi}{dx}$ are continuous at the boundary.

For infinite potential, the continuity of wave function is good enough, as wave function vanishes across boundary.

Values along the
boundary curve

$$\psi(0) = \psi(a) = 0,$$

The wave function inside the well
matches to the values outside

$$\psi(0) = A \sin 0 + B \cos 0 = B,$$

$$\rightarrow B = 0$$

$$\psi(x) = A \sin kx.$$

$$\rightarrow \psi(a) = A \sin ka = 0$$

A=0 makes the whole
wave function zero.

$$\sin ka = 0,$$

$$\rightarrow ka = 0, \pm\pi, \pm2\pi, \pm3\pi, \dots$$

But $k=0$ makes the wave function vanish, and negative
solutions can be adjusted to A following $\sin(-x)=-\sin(x)$.

$$k_n = \frac{n\pi}{a}, \quad \text{with } n = 1, 2, 3, \dots$$

$$\rightarrow E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.$$

In sharp contrast to the
classical case, the quantum
particle trapped inside can
only possess some specific
discrete energy values.

Region governed
by differential
equation.

Normalisation of wave function

$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1 \quad \rightarrow \quad \int_0^a |A|^2 \sin^2(kx) dx = |A|^2 \frac{a}{2} = 1, \quad \text{so} \quad |A|^2 = \frac{2}{a}.$$

$\psi(x) = 0$
At and beyond
 $x = \{0, a\}$

It is simplest to pick up the positive real root as the phase of A carries no physical meaning

$$\therefore A = \sqrt{2/a}$$

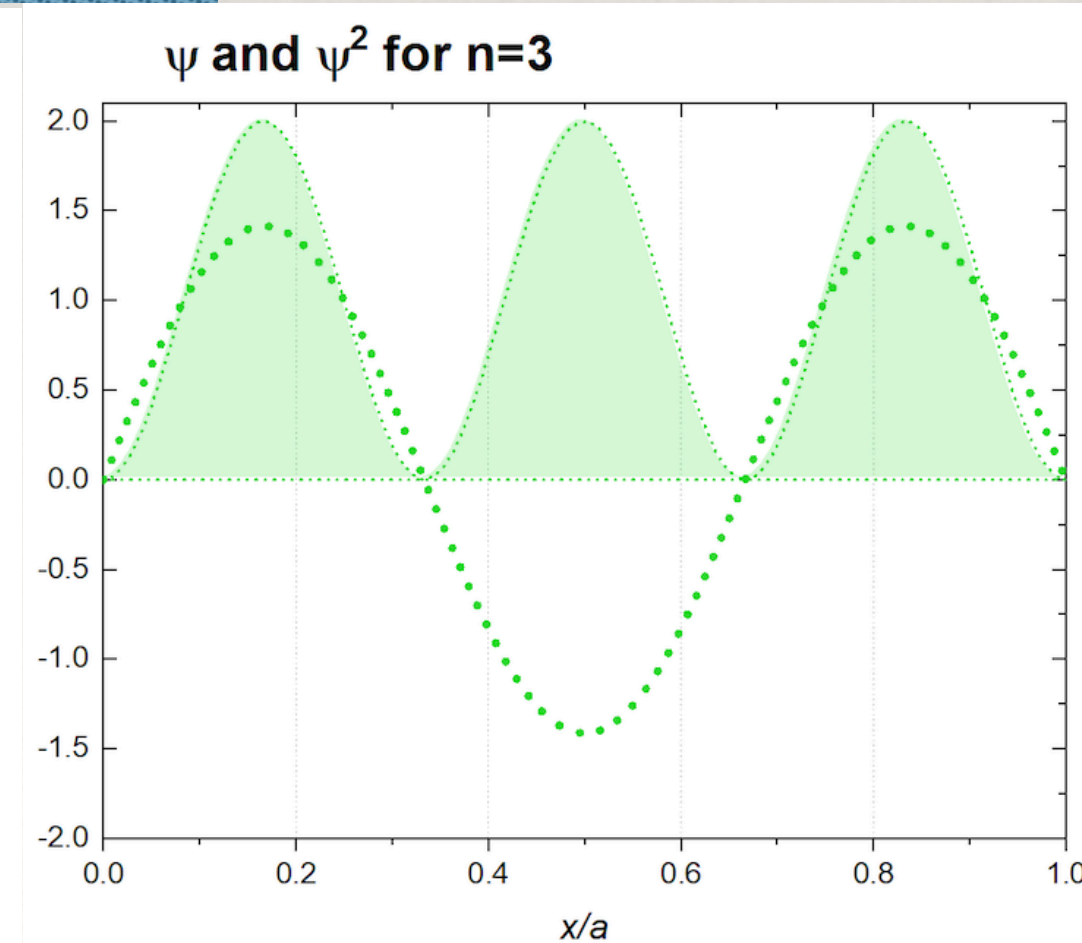
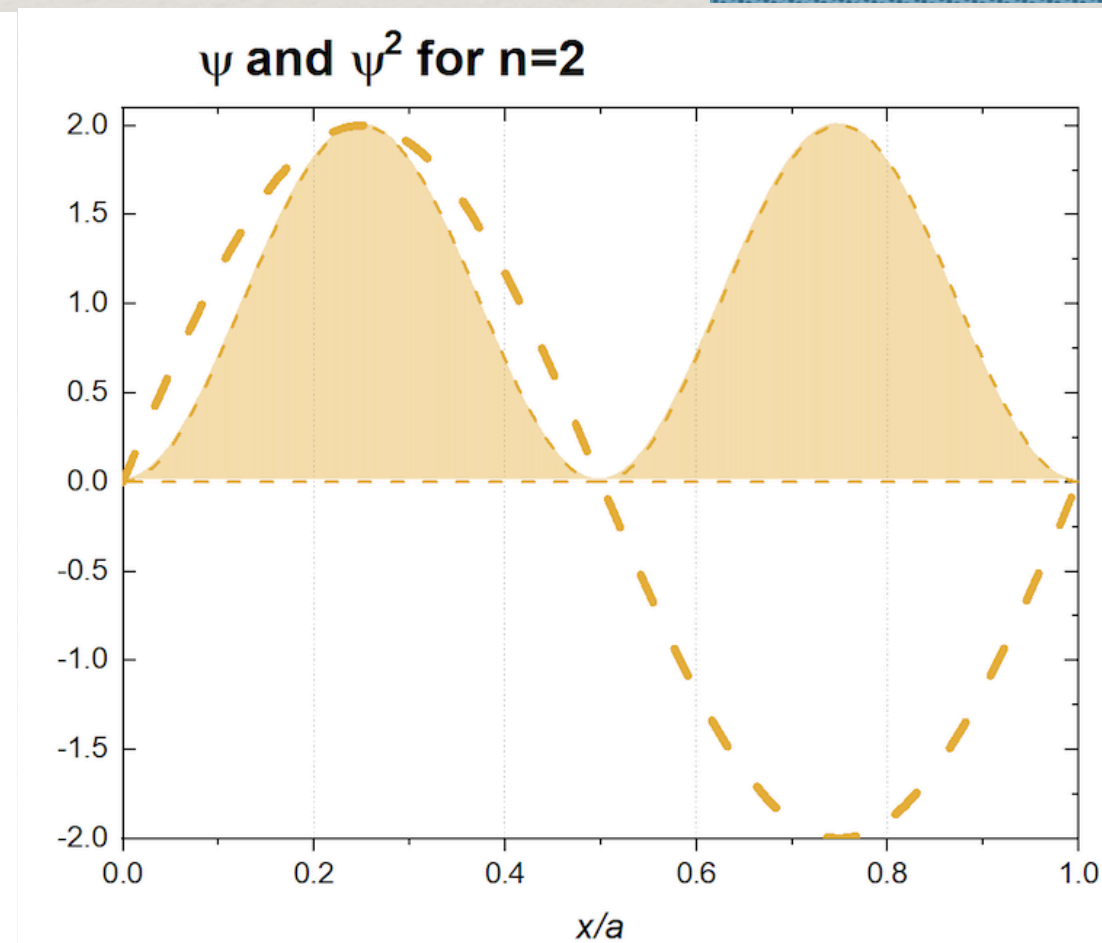
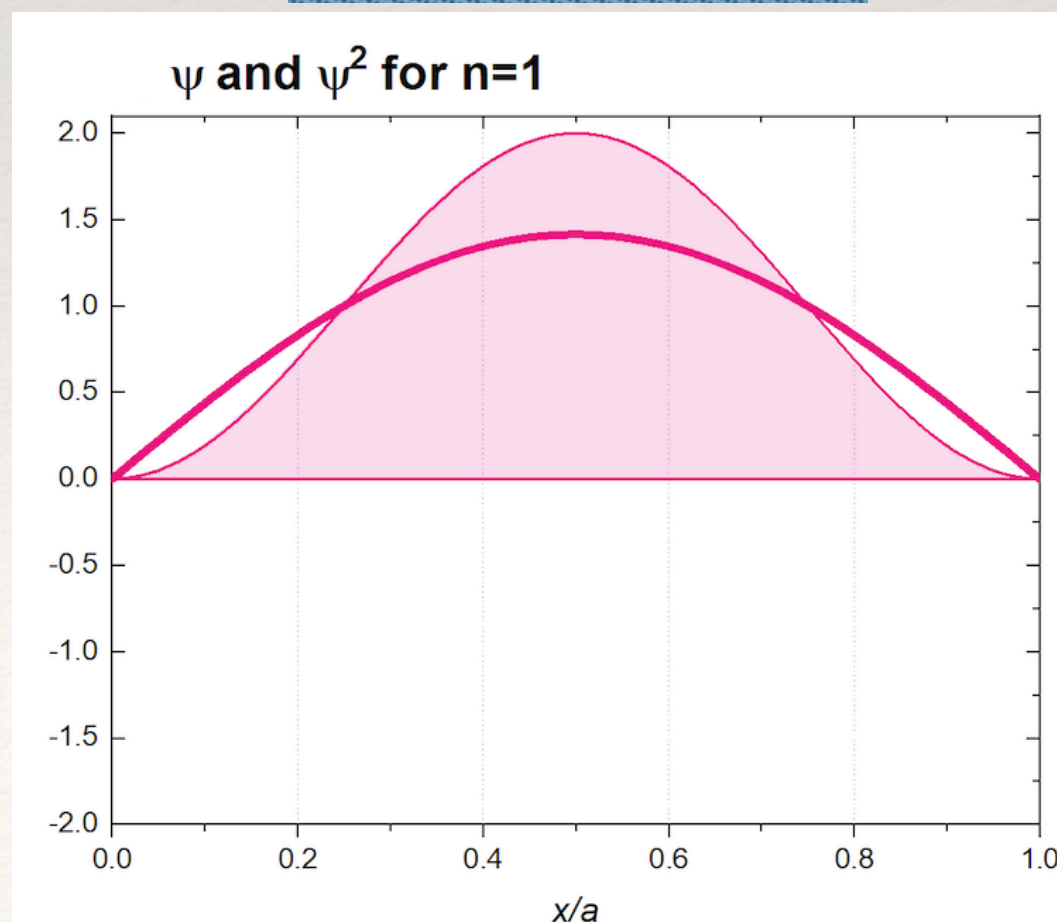
$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right).$$

$$n = 1, 2, 3 \dots$$

As promised, the solution to time independent Schrodinger Equation has provided infinitely many solutions.

Ground state

Excited states



- Each successive states add one extra node as we go up the energy.

- Ground state energy is non-zero, unlike the classical case

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$$

Generic Solutions to Schrodinger Equation

Recall, we obtained,
Solution of the form,
Called **Stationary State**

$$\Psi(x, t) = \psi(x)e^{-iEt/\hbar},$$

However, is the solution unique ?

Actually, one obtains infinitely many such solutions!

$$\begin{aligned}\psi_1(x) &\rightarrow E_1 \\ \psi_2(x) &\rightarrow E_2 \\ \psi_3(x) &\rightarrow E_3 \\ &\vdots\end{aligned}$$

Each
With
Different
Eigenvalues

}

$$\begin{aligned}\Psi_1(x, t) &= \psi_1(x)e^{-iE_1t} \\ \Psi_2(x, t) &= \psi_2(x)e^{-iE_2t} \\ \Psi_3(x, t) &= \psi_3(x)e^{-iE_3t} \\ &\vdots\end{aligned}$$

One can easily check that any linear combination of the solution to the (time dependent) Schrodinger Equation is also a solution.

The generic solution is
then a linear combination:

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

One needs to find the
unknown constant 'c'
from initial conditions

Orthogonality/Orthonormality

We will now show that there is an important property related to the wave functions:

$$\int \psi_m(x)^* \psi_n(x) dx = 0,$$

whenever $m \neq n$.

$$\int \psi_m(x)^* \psi_n(x) dx = \frac{2}{a} \int_0^a \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{a}x\right) dx$$

$$= \frac{1}{a} \int_0^a \left[\cos\left(\frac{m-n}{a}\pi x\right) - \cos\left(\frac{m+n}{a}\pi x\right) \right] dx$$

$$= \left\{ \frac{1}{(m-n)\pi} \sin\left(\frac{m-n}{a}\pi x\right) - \frac{1}{(m+n)\pi} \sin\left(\frac{m+n}{a}\pi x\right) \right\} \Big|_0^a$$

$$= \frac{1}{\pi} \left\{ \frac{\sin[(m-n)\pi]}{(m-n)} - \frac{\sin[(m+n)\pi]}{(m+n)} \right\} = 0.$$

However, when $m=n$, this integral yields 1 and thus provides normalisation. Check !

Recall that unit vectors also satisfy such orthonormality conditions.
 $i \cdot j = i \cdot k = j \cdot k = 0$, $i \cdot i = j \cdot j = k \cdot k = 1$

Orthonormality

$$\int \psi_m(x)^* \psi_n(x) dx = \delta_{mn},$$

$$\delta_{mn} = \begin{cases} 0, & \text{if } m \neq n; \\ 1, & \text{if } m = n. \end{cases}$$

Kronecker Delta function

Completeness

$\psi_n(x)$ are *complete* set of functions.

$$\Psi(x) = \sum_{n=1}^{\infty} c_n \psi_n(x)$$

A generic solution can be expressed as a linear combinations of them.

$$= \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi}{a}x\right).$$

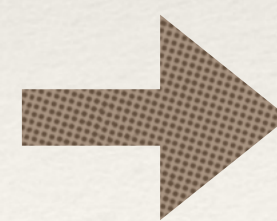
Its a Fourier series for a function.

The expansion coefficients can be found out by Fourier's trick!

Using orthonormalisation!

Consider, $\int \psi_m^*(x) \Psi(x) dx = \sum_{n=1}^{\infty} c_n \int \psi_m(x)^* \psi_n(x) dx = \sum_{n=1}^{\infty} c_n \delta_{mn} = c_m.$

Kronecker Delta kills every term excepting $n=m$



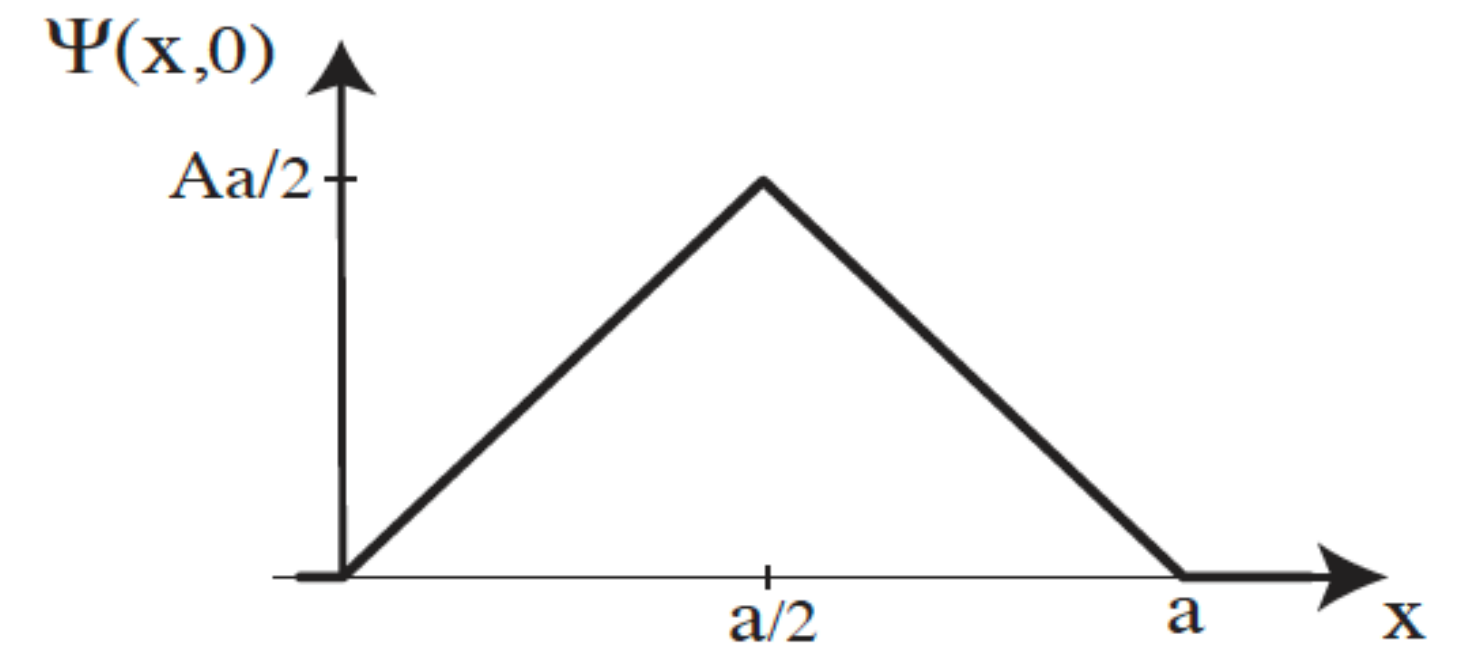
$$c_m = \int \psi_m^*(x) \Psi(x) dx$$

Given the wave function at $t=0$, we can find the complete solution.

Example

For a particle
in an infinite
potential well

$$\begin{aligned}\Psi(x, 0) &= Ax \quad \text{for } 0 \leq x \leq a/2; \\ &= A(a - x) \quad \text{for } a/2 \leq x \leq a.\end{aligned}$$



$$\begin{aligned}1 &= A^2 \int_0^{a/2} x^2 dx + A^2 \int_{a/2}^a (a - x)^2 dx = A^2 \left[\frac{x^3}{3} \Big|_0^{a/2} - \frac{(a - x)^3}{3} \Big|_{a/2}^a \right] \\ &= \frac{A^2}{3} \left(\frac{a^3}{8} + \frac{a^3}{8} \right) = \frac{A^2 a^3}{12} \Rightarrow \boxed{A = \frac{2\sqrt{3}}{\sqrt{a^3}}}.\end{aligned}$$

$$\begin{aligned}c_n &= \sqrt{\frac{2}{a}} \frac{2\sqrt{3}}{a\sqrt{a}} \left[\int_0^{a/2} x \sin\left(\frac{n\pi}{a}x\right) dx + \int_{a/2}^a (a - x) \sin\left(\frac{n\pi}{a}x\right) dx \right] \\ &= \frac{2\sqrt{6}}{a^2} \left\{ \left[\left(\frac{a}{n\pi}\right)^2 \sin\left(\frac{n\pi}{a}x\right) - \frac{xa}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \right] \Big|_0^{a/2} \right. \\ &\quad \left. + a \left[-\frac{a}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \right] \Big|_{a/2}^a - \left[\left(\frac{a}{n\pi}\right)^2 \sin\left(\frac{n\pi}{a}x\right) - \left(\frac{ax}{n\pi}\right) \cos\left(\frac{n\pi}{a}x\right) \right] \Big|_{a/2}^a \right\} \\ &= \frac{2\sqrt{6}}{a^2} \left[\left(\frac{a}{n\pi}\right)^2 \sin\left(\frac{n\pi}{2}\right) - \frac{a^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{a^2}{n\pi} \cos n\pi + \frac{a^2}{n\pi} \cos\left(\frac{n\pi}{2}\right) \right. \\ &\quad \left. + \left(\frac{a}{n\pi}\right)^2 \sin\left(\frac{n\pi}{2}\right) + \frac{a^2}{n\pi} \cos n\pi - \frac{a^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) \right]\end{aligned}$$

$$c_n = \frac{2\sqrt{6}}{a^2} 2 \frac{a^2}{(n\pi)^2} \sin\left(\frac{n\pi}{2}\right) = \frac{4\sqrt{6}}{(n\pi)^2} \sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0, & n \text{ even,} \\ (-1)^{(n-1)/2} \frac{4\sqrt{6}}{(n\pi)^2}, & n \text{ odd.} \end{cases}$$

$$\Psi(x, t) = \frac{4\sqrt{6}}{\pi^2} \sqrt{\frac{2}{a}} \sum_{n=1,3,5,\dots} (-1)^{(n-1)/2} \frac{1}{n^2} \sin\left(\frac{n\pi}{a}x\right) e^{-E_n t/\hbar},$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.$$

Summary

- When the potential is time independent, the Schrodinger Equation can be solved using the method of separation of variables.

$$\Psi(x, t) = \psi(x)f(t)$$

- Separable solutions provide **stationary states**, for which probability density and expectation values are time independent.
- Infinite potential well provides discrete energy Eigen values or Eigen states where the particles have definite energy and no uncertainty. The eigenstates are orthonormal.

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right).$$

$$n = 1, 2, 3...$$

- The generic solution is the superposition of all these eigenstate solutions with arbitrary coefficients.

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-i E_n t / \hbar}.$$

- The coefficients can be found out if the initial wave function of the system is known.